



NEXA PLATFORM

Enterprise SaaS Release Notes & System Changelog

Release Metadata

Version:	v4.8.0
Date:	July 24, 2026
Track:	General Availability
Build:	#nexa-4.8.0-r89412

1. Executive Summary & Highlights

Q3 Infrastructure Milestone

Nexa Platform Version 4.8.0 represents our most significant architectural update of Q3 2026. This release delivers our next-generation Real-Time Streaming Analytics Engine (v2), Zero-Trust Automated API Key Rotation, and high-concurrency GraphQL Subscriptions over HTTP/2. Designed for high-throughput enterprise workloads, v4.8.0 improves ingestion throughput by 54% while reducing P99 API latency across global edge points of presence.

+54%

Ingestion Rate

Up to 185k events/sec

-60%

P99 API Latency

Reduced from 48ms to 19ms

99.999%

SLA Target

Multi-region active-active

0 ms

Migration Downtime

Zero-downtime schema rollout

2. Core Feature Deep Dives

Technical Architectural Upgrades

2.1 Real-Time Streaming Analytics Engine v2

The streaming engine has been re-architected around a zero-copy memory layout and asynchronous event pipelines. Built on Apache Flink and Rust-native micro-workers, Engine v2 supports dynamic partition reassignment without dropping inflight WebSocket client connections.

Key Architectural Improvements:

- Sub-millisecond State Evaluation:** In-memory sliding windows now evaluate up to 10M keys/sec per node.
- Adaptive Backpressure Management:** Automatically throttles downstream sinks during downstream cloud service degradation without failing upstream requests.
- Native Parquet Storage Sync:** Direct stream-to-S3/GCS archival with automatic snappy compression and partition pruning.

Example Streaming Analytics Pipeline Definition (nexa-stream-v2.json):

```
{
  "pipeline_id": "pl_analytics_realtime_v2",
  "source": { "provider": "kinesis", "stream_name": "nexa-events-prod" },
  "window_spec": { "type": "tumbling", "duration_seconds": 60 },
  "transformations": [
    { "type": "enrich_geoip", "cache_ttl_seconds": 3600 },
    { "type": "aggregate", "metrics": ["count", "p95_latency", "error_rate"] }
  ],
  "sink": { "target": "timescale_db", "auto_create_table": true }
}
```

2.2 Zero-Trust API Key Lifecycle & Scope Enforcer

To strengthen security compliance (SOC2 Type II, ISO 27001), Nexa v4.8.0 introduces automated short-lived API tokens and granular privilege scope enforcement. Administrators can now mandate automatic 30-day token expiration cycles with zero client disruption via grace-period overlap keys.

- Dual-Key Overlap Window:** Old API keys remain valid for a configurable grace period (e.g., 24 hours) after new key generation.
- Fine-Grained Scopes:** Over 85 granular permission strings (e.g., analytics:read, cluster:deploy, billing:admin).
- Audit Logging:** Every key generation, rotation, and revocation event is signed with SHA-256 and pushed to immutable audit ledgers.